

95 p 38

$$1) (5+2i)\bar{z} - 2 = i$$

$$\Leftrightarrow (5+2i)\bar{z} = 2+i$$

$$\Leftrightarrow \bar{z} = \frac{2+i}{5+2i}$$

$$\Leftrightarrow \bar{z} = \frac{(2+i)(5-2i)}{(5+2i)(5-2i)}$$

$$\Leftrightarrow \bar{z} = \frac{10 - 4i + 5i - 2i^2}{5^2 + 2^2}$$

$$\Leftrightarrow \bar{z} = \frac{10 + i + 2}{29}$$

$$\Leftrightarrow \bar{z} = \frac{12}{29} + \frac{1}{29}i$$

$$\Leftrightarrow z = \frac{12}{29} - \frac{1}{29}i$$

$$S_1 = \left\{ \frac{12}{29} - \frac{1}{29}i \right\}$$

$$2) 2i(1-2\bar{z}) + \bar{z} = i\bar{z} - 1$$

$$\Leftrightarrow 2i - 4i\bar{z} + \bar{z} - i\bar{z} = -1$$

$$\Leftrightarrow \bar{z}(-4i + 1 - i) = -1 - 2i$$

$$\Leftrightarrow \bar{z}(1 - 5i) = -1 - 2i$$

$$\Leftrightarrow \bar{z} = \frac{-1 - 2i}{1 - 5i}$$

$$\Leftrightarrow \bar{z} = \frac{(-1 - 2i)(1 + 5i)}{(1 - 5i)(1 + 5i)}$$

$$\Leftrightarrow \bar{z} = \frac{-1 - 5i - 2i - 10i^2}{1^2 + 5^2}$$

$$\Leftrightarrow \bar{z} = \frac{9 - 7i}{26}$$

$$\Leftrightarrow z = \frac{9}{26} + \frac{7}{26}i$$

$$S_2 = \left\{ \frac{9}{26} + \frac{7}{26}i \right\}$$

$$3) 2i\bar{z} - i = 2(\bar{z} - 5) + i \quad \Leftrightarrow \bar{z} = \frac{(-5+i)(-1-i)}{(-1+i)(-1-i)}$$

$$\Leftrightarrow 2i\bar{z} - i = 2\bar{z} - 10 + i \quad \Leftrightarrow \bar{z} = \frac{5+5i-i-i^2}{(-1)^2+1^2}$$

$$\Leftrightarrow 2i\bar{z} - 2\bar{z} = -10 + 2i \quad \Leftrightarrow \bar{z} = \frac{6+4i}{2}$$

$$\Leftrightarrow \bar{z}(-2+2i) = -10+2i \quad \Leftrightarrow \bar{z} = 3+2i$$

$$\Leftrightarrow \bar{z} = \frac{-10+2i}{-2+2i}$$

$$\Leftrightarrow \bar{z} = 3-2i$$

$$\Leftrightarrow \bar{z} = \frac{-5+i}{-1+i}$$

$$\underline{S_3 = \{3-2i\}}$$

4) Pour $\bar{z} \neq 1$ (c'est à dire: $z \neq 1$):

$$\frac{\bar{z}-2i}{\bar{z}-1} + 1+i = 0$$

$$\Leftrightarrow \bar{z} = 1+i$$

$$\Leftrightarrow \bar{z} = 1-i$$

$$\Leftrightarrow \bar{z} - 2i + (1+i)(\bar{z}-1) = 0$$

$$\underline{S_4 = \{1-i\}}$$

$$\Leftrightarrow \bar{z} - 2i + (1+i)\bar{z} - 1 - i = 0$$

$$\Leftrightarrow \bar{z}(2+i) - 1 - 3i = 0$$

$$\Leftrightarrow \bar{z} = \frac{1+3i}{2+i}$$

$$\Leftrightarrow \bar{z} = \frac{(1+3i)(2-i)}{(2+i)(2-i)}$$

$$\Leftrightarrow \bar{z} = \frac{2-i+6i-3i^2}{2^2+1^2}$$

$$\Leftrightarrow \bar{z} = \frac{5+5i}{5}$$